

COMS 6113

Agentic Systems

Fall 2026, Friday 10:10-12pm, 227 Mudd

daplab.cs.columbia.edu/agentic-systems

Who are we?

Kostis Kaffes kkaffes@cs.columbia.edu

Assistant professor in the CS Department working on Systems + Agents

Earlier life: Low latency systems and networks for the cloud

Georgios Liargkovas gl2902@columbia.edu

3rd-year PhD in the CS department working with Kostis

Research mostly focused on Agents *for* Systems

Who are you?

Agentic (Computer) Systems

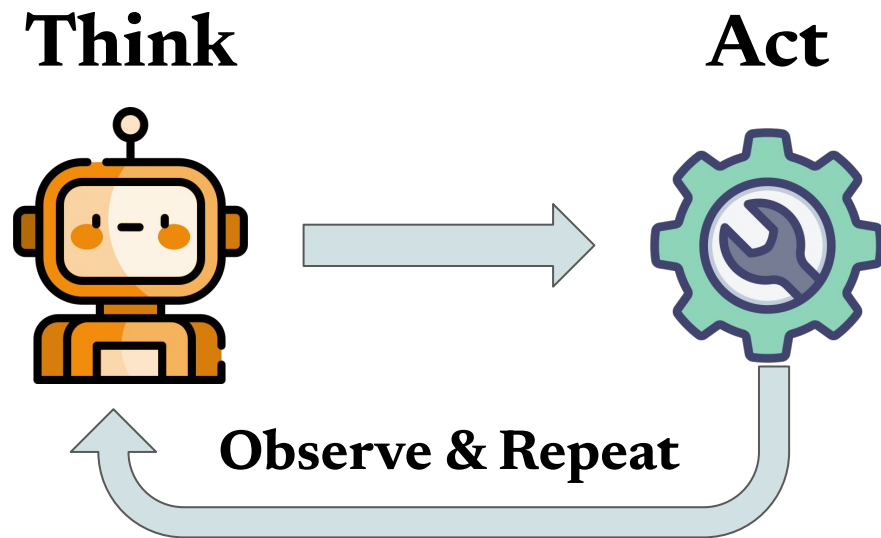
- Agents for Systems
- Systems for Systems

We want to explore many different aspects:

- Make {agents, systems} more efficient.
- Make {agents, systems} faster.
- Make {agents, systems} safer/more secure.
- Make agents more capable.

What is an agent?

What is an agent?



Class Format

Class built around a semester-long research project.

End of Semester

- Final deliverable is a *paper* ready to be submitted to top-tier conference.
- You will review each other's papers.
- Final presentations on 12/11 (*you need to attend*)

Class Format

During the semester

- All students actively participate and present progress *every* week.
- Lectures will be split across project updates - research methodology - relevant work (papers).
- Each group will lead a lecture on their related work randomly assigned between 10/9-11/13 (*you can swap lectures - let us know early!*)
- Attendance is mandatory + no open laptops allowed.
- All announcements/discussions happen on Slack (no emails!) – **You are responsible to monitor it.**

Warning: This will be a demanding class!

Grading

25% Class participation / discussion

25% Weekly updates + related work lecture

25% Final presentation

25% Final paper + reviewing

Project

Work on a predefined project* – you cannot bring your own PhD/hobby/start-up idea to the class (will go over possible projects later).

Groups of ~4 – minimum 2.

Target outcome is an OSDI/ICML/VLDB submission.

Weekly project updates.

Many intermediate deliverables.

*** You can bring your own project if:**

1. You send it to me **TODAY**.
2. It is about agents + systems.
3. You are okay with more people joining your project.
4. I agree with it. :)

Course website + Schedule

Immediate TODOs

1. Get to now and chat with each other after class today.
2. Sign up for the course [Slack](#).
3. Read the [project descriptions](#).
4. *By EOD Monday:*
 - a. Discuss with other students on Slack.
 - b. Decide on a project.
 - c. Tag Kostis and Georgios on the Slack channel with the team.
 - d. After that *please* do not drop the course.

Waitlist?

WHOEVER HAS:

- 1) **DECIDED ON A PROJECT.**
- 2) **COMMUNICATED THE DECISION TO US THROUGH SLACK IN THE PROJECT'S CHANNEL.**

BY EOD MONDAY

WILL BE ADMITTED TO THE CLASS FROM THE WAITLIST

Next few weeks

09/18 – Project pitch

09/25 – Short overview of related work by every team

Write up related work for the paper

10/02 – No class

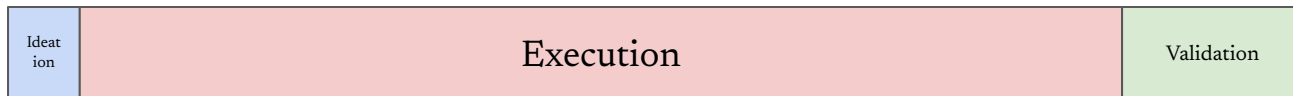
10/09 – Run a baseline, show us the first figure

Due 09/18: Pitch aka The Heilmeier Catechism

1. What are you trying to do? Articulate your objectives using absolutely no jargon.
2. How is it done today, and what are the limits of current practice?
3. What is new in your approach and why do you think it will be successful?
4. Who cares? If you are successful, what difference will it make?
5. What are the risks?
- ~~6. How much will it cost?~~
- ~~7. How long will it take? One semester!~~
8. What are the midterm and final “exams” to check for success?

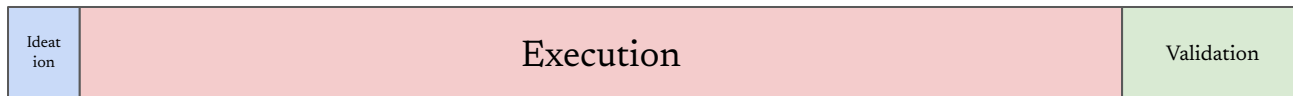
Research and AI Agents

Math Before



Research and AI Agents

Math Before

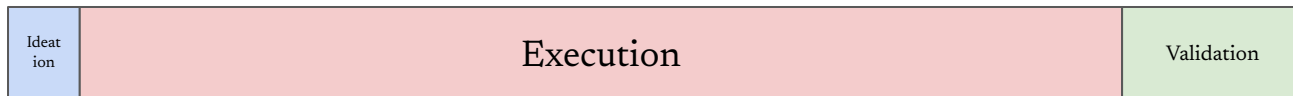


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Research and AI Agents

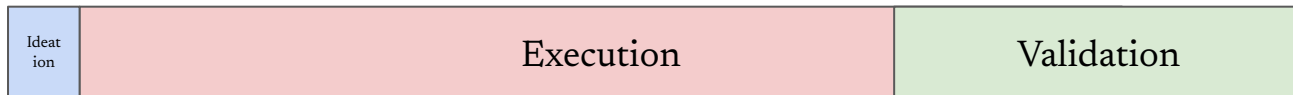
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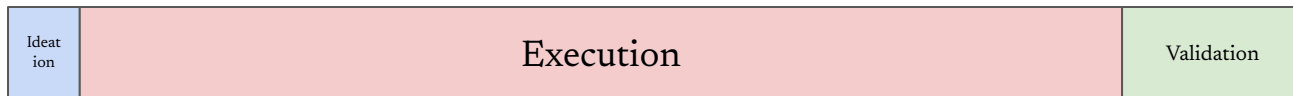


Biology Before



Research and AI Agents

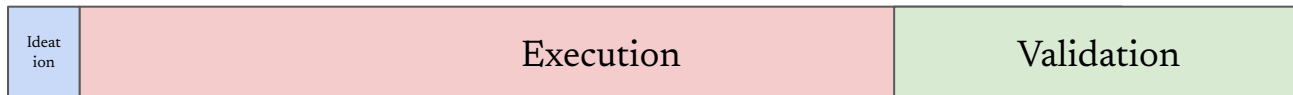
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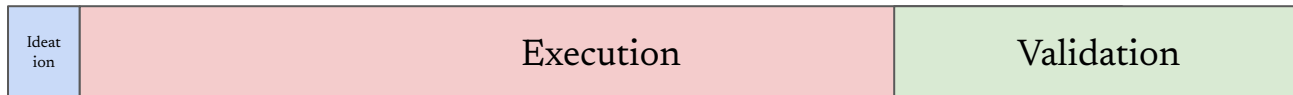
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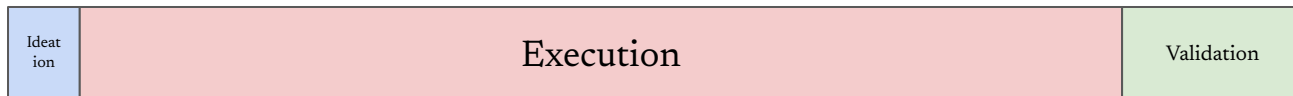


Biology After



Research and AI Agents

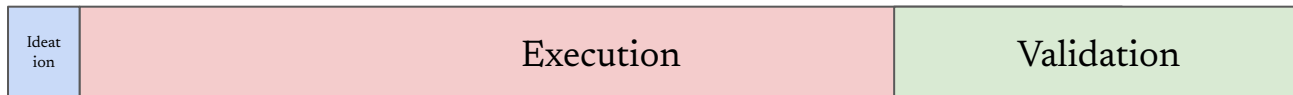
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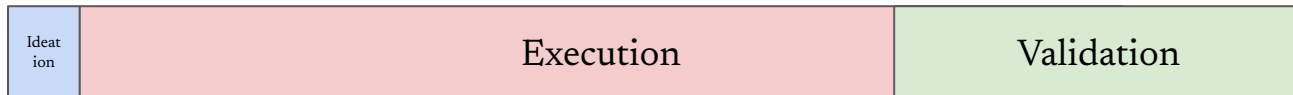
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Biology Before



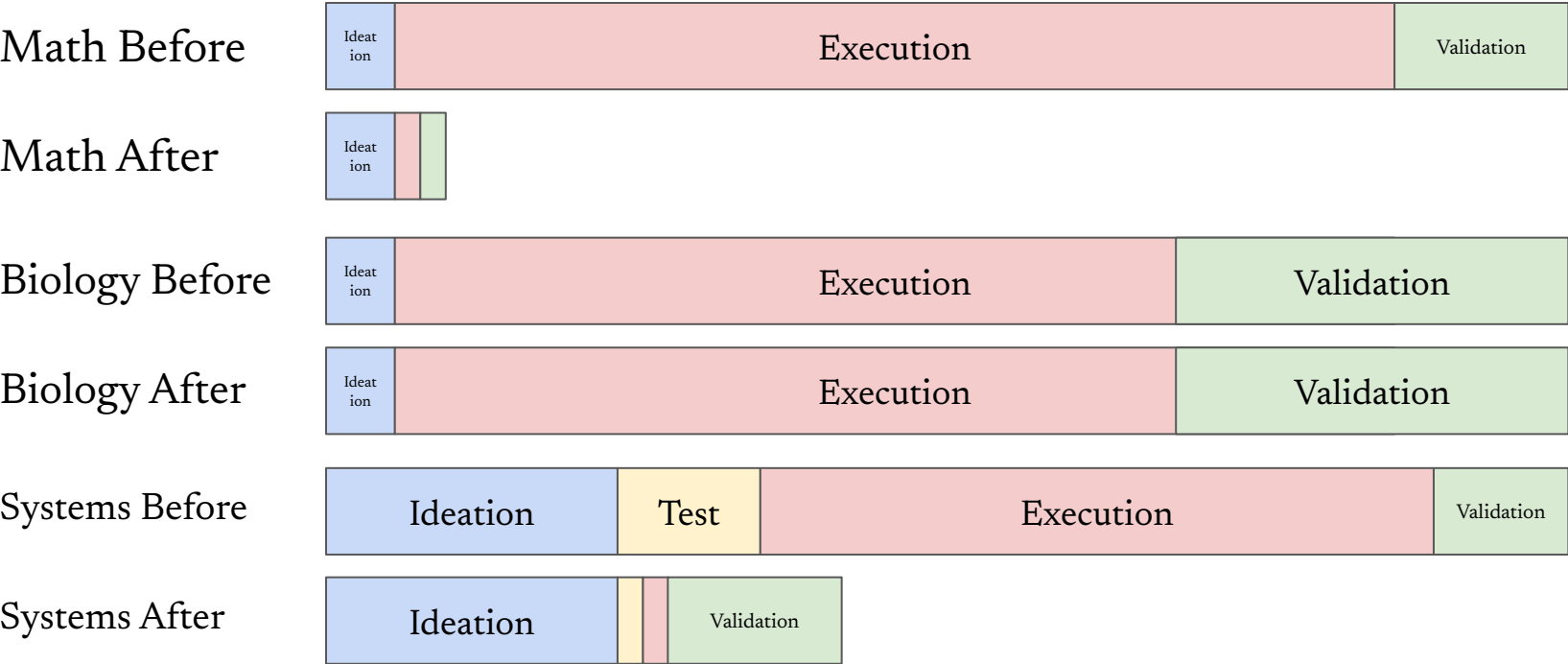
Biology After



Systems Before



Research and AI Agents



How to be a great researcher

- **Knowledge compounds**
 - *“Luck favors the prepared mind” – Pasteur*
- **Be courageous**
 - *You need to solve something no-one has solved before*
- **Tolerate ambiguity**
 - *If it was clear, it would have been solved*
- **Go wide**
 - *You will get ideas from other areas*
- **Go deep**
 - *Understand in depth, don't just let the LLM read the paper for you*

How to be a great researcher

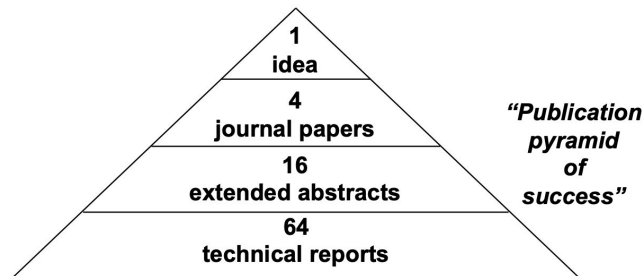
- **When an opportunity opens up, get after it and pursue it**
 - *People who switched focus to LLMs in 2022 were not disappointed*
- **Who you work with matters more than what you work on exactly**
 - *Choose the right people to bounce ideas off of*
- **Do your job in such a fashion that others can build on it**
 - *If no-one uses your work, it has zero impact*
- **Need to sell your work, via good writing and good talks**
 - *Make all your talks for a general audience*
- ???

How to be a bad researcher

Dave Patterson
inventor of RISC
Turing Award



1. **Be THE leading expert**
2. **Let Complexity Be Your Guide**
3. **Never be Proven Wrong**
 - Avoid {implementing, testing, benchmarking}
4. **Don't be Distracted by Others (Avoid Feedback)**
5. **Papers: It's Quantity, not Quality**



Projects